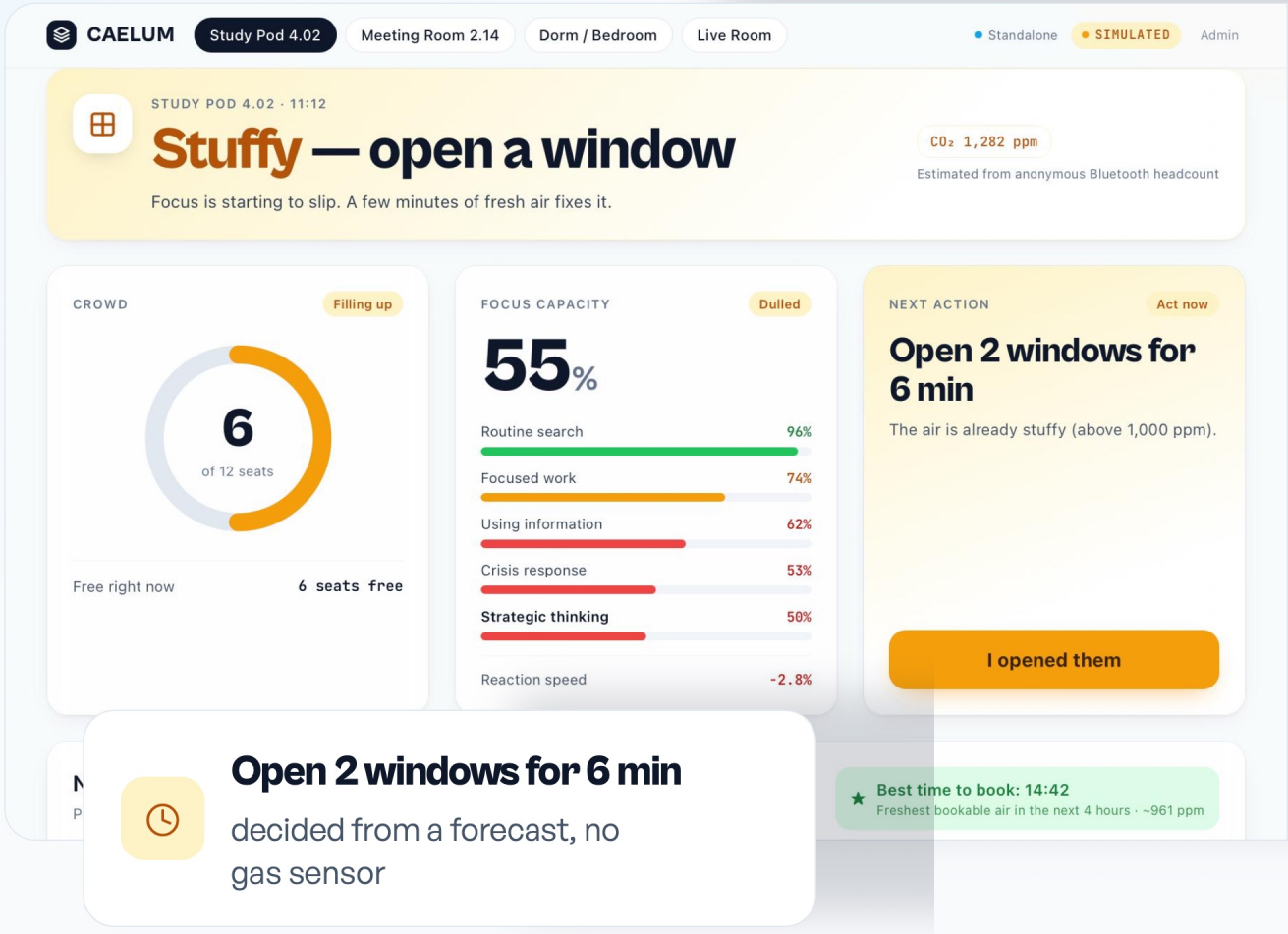
VIRTUAL CO_2 SENSOR · LIVE DEMO

Predict the air. Protect the focus.

Caelum turns anonymous Bluetooth headcounts into a 4-hour CO_2 forecast, then tells you exactly when to open the window and for how long.

- Anonymous Bluetooth
- CO_2 physics
- Harvard COGfx



Why the air is worth predicting

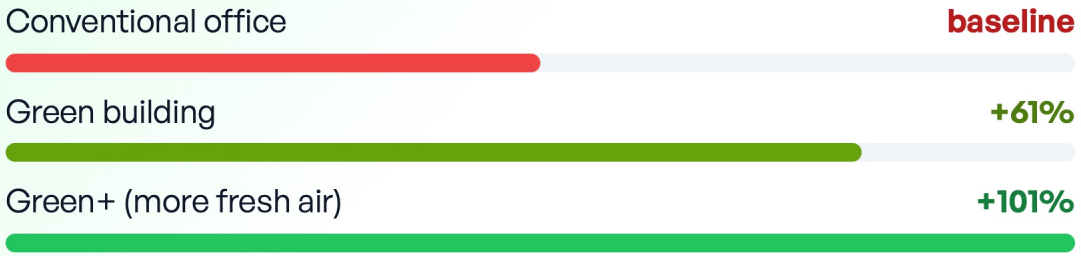
What controlled studies measured, and what a forecast lets you act on.

COGNITIVE FUNCTION

Harvard COGfx 2016

+101%

higher cognitive scores with more fresh air



Allen 2016

−21%

cognitive score for every extra 400 ppm of CO_2

COGfx Global 2021

−2.4%

correct answers per minute, per +500 ppm

LBNL 2012

1,000 ppm

is where decision-making starts to slip

MacNaughton 2015

$6,500 / worker

a year in productivity, for under $40 of air

VIRTUAL SENSORS

Nothing to mount

Cheaper wherever an estimate is enough

One reusable Bluetooth scanner per area replaces a wired CO_2 sensor in every room: no install, no calibration, no removal.

DTU 2016

Better sleep

Lower bedroom CO_2, sharper next-day thinking

4 h ahead

Act early

The forecast flags stuffy air before it arrives

From live signal to the right decision

Alternative data in, one decision out.

01 Sense
Scanners count phones in the room by Bluetooth

→

02 Forecast
Models predict who will be in, next 24 hours

→

03 Simulate
Mass-balance physics turns people into ppm

→

04 Decide
When to open which windows, and for how long

Live

Bluetooth headcount

Counts phones in range, skips earbuds and trackers.
No cameras.

Model

Occupancy forecast

Gradient boosting vs. ridge, seasonal and persistence, best per horizon.

Model

Honest uncertainty

80% bands; 0.05–0.3 people mean error in blind backtests.

Physics

Virtual CO_2 sensor

Physics turns people, room volume and windows into ppm.

COGfx

Focus capacity

Harvard COGfx curve: focus % per task, from routine to strategy.

Act now

Ventilation timing

“Open 2 windows for 6 min”: when, how many and how long.

Best time

Smart booking

Best time to book, plus 10-min fresh-air breaks between meetings.

Admin

Ops console

Tune scanner range, watch heartbeats, see where people sit.